
Gama Cirques: Decoding an Olympic Pandiagonal Magic Square

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Abstract

Gama Cirques is Puzzle 2 from the GRiddles Riddles Series A. The title phrase is an anagram of MAGIC SQUARE, and the puzzle image shows a 4×4 grid of coloured Olympic rings bearing six text labels in as many distinct calendar systems. Each label encodes a specific Summer Olympics; together with the remaining visually identified Games, the sixteen IOC editions from Rome 1960 through Tokyo 2020 (ordinals 17–32) are placed in the grid. These ordinals form a pandiagonal magic square with magic constant $M = 98$. One cell carries only the symbol “?”; the magic square constraints uniquely force that cell to equal ordinal 26, corresponding to the 1996 Atlanta Summer Olympics. The softball gold medal at Atlanta 1996 was won by

The United States.

1 Introduction

The GRiddles Riddles Series is a collection of layered logic puzzles in which each puzzle encodes its answer behind several independent mechanisms. Puzzle 2, titled *Gama Cirques*, presents the solver with a 4×4 arrangement of Olympic-ring symbols, each ring coloured in one of four Olympic colours (blue, yellow, red, or green). Six rings contain text labels; one ring contains the symbol “?”; the remaining nine are empty rings. The question posed is: *Who was the softball champion?*

Figure 1 shows the original puzzle image. Three nested layers lead to the answer:

1. **Anagram.** The title “Gama Cirques” is an anagram of MAGIC SQUARE.
2. **Olympic identification.** The coloured rings represent Summer Olympics, and the six text labels encode specific editions via six distinct calendar systems.
3. **Magic square.** The decoded IOC ordinal numbers fill a pandiagonal magic square; the mystery cell is the unique ordinal completing it.

2 Layer 1: The Anagram

The phrase “Gama cirques” contains eleven letters (ignoring the space):

G, a, m, a, a, c, i, r, q, u, e, s.

Sorted alphabetically: a, a, c, e, g, i, m, q, r, s, u. The phrase MAGIC SQUARE contains the same eleven letters in the same multiset:

m, a, g, i, c, s, q, u, a, r, e $\longrightarrow$ a, a, c, e, g, i, m, q, r, s, u.

The title is therefore an exact anagram of MAGIC SQUARE. This instructs the solver to look for a magic square structure in the 4×4 grid.

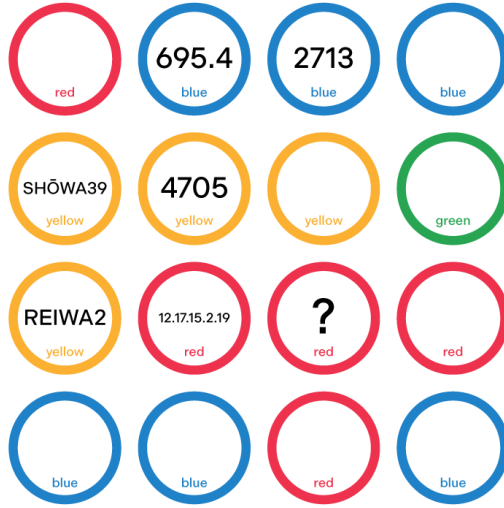


Figure 1: The original Puzzle 2 image. A 4×4 grid of coloured Olympic rings: six carry calendar-system labels, one carries “?”, and nine are empty.

3 Layer 2: Olympic Identification

3.1 Ring colours and continents

The five Olympic ring colours encode the five inhabited continental regions:

Ring colour	Continental region
Blue	Europe
Yellow	Asia
Red	Americas
Green	Oceania
Black	Africa

The 4×4 grid contains 16 rings in four colours; black is absent because no African city hosted the Summer Olympics during the relevant period.

3.2 Identifying the sixteen editions

The sixteen Summer Olympics from Rome 1960 through Tokyo 2020 (IOC ordinals 17–32) are exactly the Games hosted on those four continents. Nine rings are identified directly by their colour and the city visible from context; six carry explicit calendar encodings (Table 1).

The nine visually-identified editions (colour context and known host-city knowledge) fill the remaining cells. Table 2 lists all sixteen assignments.

4 Layer 3: The Pandiagonal Magic Square

4.1 Magic constant

The sixteen IOC ordinals 17 through 32 are used exactly once each. Their sum is

$$\sum_{k=17}^{32} k = \frac{(17 + 32) \cdot 16}{2} = 392.$$

Table 1: Calendar-system encodings on the six labelled rings. Each label uniquely identifies a Summer Olympics.

Label	Calendar system	Conversion	Year	IOC #
695.4	Greek Olympiads	Olympiad 695, year 4 $\Rightarrow$ 2004 CE	2004	28
2713	Roman AUC	$2713 - 753 = 1960$	1960	17
SHŌWA 39	Japanese Shōwa era	$1925 + 39 = 1964$	1964	18
4705	Chinese Huangdi	$4705 - 2697 = 2008$	2008	29
REIWA 2	Japanese Reiwa era	$2018 + 2 = 2020$	2020	32
12.17.15.2.19	Mayan Long Count	October 1968 CE	1968	19

Table 2: The sixteen Summer Olympics and their grid positions. Shading indicates continental region: blue (Europe), yellow (Asia), red (Americas), green (Oceania). The mystery cell is marked “?”.

	Col 0	Col 1	Col 2	Col 3
Row 0	23 LA '84	28 Athens '04	17 Rome '60	30 London '12
Row 1	18 Tokyo '64	29 Beijing '08	24 Seoul '88	27 Sydney '00
Row 2	32 Tokyo '20	19 Mexico City '68	?	21 Montréal '76
Row 3	25 Barcelona '92	22 Moscow '80	31 Rio '16	20 Munich '72

For a 4×4 magic square to have all rows summing equally, the magic constant must be

Proposition 4.1. $M = 392/4 = 98$.

Every row, every column, and every main diagonal of the decoded grid (Figure 2) sums to 98. One can verify directly:

$$\text{Rows: } 23+28+17+30 = 18+29+24+27 = 32+19+?+21 = 25+22+31+20 = 98,$$

$$\text{Columns: } 23+18+32+25 = 28+29+19+22 = 17+24+?+31 = 30+27+21+20 = 98,$$

$$\text{Diagonals: } 23+29+?+20 = 30+24+19+25 = 98.$$

All three constraints are satisfied when and only when $? = 26$.

4.2 Pandiagonal and complement-pair properties

The square is in fact *pandiagonal*: all four broken diagonals in each direction also sum to M . Denoting the matrix as A , with rows and columns indexed 0–3 modulo 4, the broken diagonals are:

$$A_{0,1} + A_{1,2} + A_{2,3} + A_{3,0} = 28 + 24 + 21 + 25 = 98,$$

$$A_{0,2} + A_{1,3} + A_{2,0} + A_{3,1} = 17 + 27 + 32 + 22 = 98,$$

$$A_{0,3} + A_{1,0} + A_{2,1} + A_{3,2} = 30 + 18 + 19 + 31 = 98,$$

and symmetrically for the other direction.

Additionally, the square satisfies the *complement-pair property*: every cell and the cell offset from it by $(+2, +2) \pmod{4}$ sum to $M/2 = 49$. For instance, $A_{0,0} + A_{2,2} = 23 + 26 = 49$. All eight such pairs are:

Positions	Values	Sum
$(0,0) \leftrightarrow (2,2)$	$23 + 26$	49
$(0,1) \leftrightarrow (2,3)$	$28 + 21$	49
$(0,2) \leftrightarrow (2,0)$	$17 + 32$	49
$(0,3) \leftrightarrow (2,1)$	$30 + 19$	49
$(1,0) \leftrightarrow (3,2)$	$18 + 31$	49
$(1,1) \leftrightarrow (3,3)$	$29 + 20$	49
$(1,2) \leftrightarrow (3,0)$	$24 + 25$	49
$(1,3) \leftrightarrow (3,1)$	$27 + 22$	49

Furthermore, every toroidal 2×2 block sums to $M = 98$. These three properties — pandiagonality, the complement-pair property, and the 2×2 block property — together characterise a *Parshvanath*-type most-perfect magic square, a structure known in Indian combinatorics for over a millennium.

Decoded 4×4 Parshvanath Pandiagonal Magic Square ($M = 98$)



Figure 2: The decoded 4×4 pandiagonal magic square. Cell colour indicates continental region (red = Americas, blue = Europe, yellow = Asia, green = Oceania). Cells with calendar-system encodings show the encoded label in italic. The mystery cell is marked with a thick black border and revealed as ordinal 26 (Atlanta 1996). All rows and columns sum to $M = 98$ (edge annotations); both main diagonals (dashed lines) also sum to 98.

5 Identifying the Mystery Cell

Three independent solve paths all yield $? = 26$.

Path A (row/column linear system). Row 2 gives $32 + 19 + ? + 21 = 98$, so $? = 98 - 72 = 26$. Column 2 independently gives $17 + 24 + ? + 31 = 98$, so $? = 98 - 72 = 26$. Both diagonals also constrain to 26.

Path B (exhaustive ordinal check). The sixteen ordinals 17–32 must appear exactly once. Those appearing in the grid are {17, 18, 19, 20, 21, 22, 23, 24, 25, 27, 28, 29, 30, 31, 32}. The unique missing ordinal is 26.

Path C (complement-pair). Los Angeles 1984 occupies position (0, 0), ordinal 23. Its complement-pair partner at (2, 2) must sum to $M/2 = 49$, so $? = 49 - 23 = 26$.

6 The Answer

The mystery cell is IOC ordinal 26, which identifies the **1996 Atlanta Summer Olympics**. The puzzle asks for the softball champion at those Games. Women’s softball made its Olympic debut at Atlanta 1996; the gold medal was won by

The United States.

Remark 6.1. *The magic constant $M = 98$ and its half $M/2 = 49$ appear as exponents in the closed-form answer to Puzzle 1 of this same series: $3(2^{198} + 2^{98} + 2^{49})$. The mystery ordinal 26 also matches the last two digits of the current year (2026). Whether these are deliberate cross-puzzle encodings or numerical coincidences is left to the reader.*

Acknowledgements

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References

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